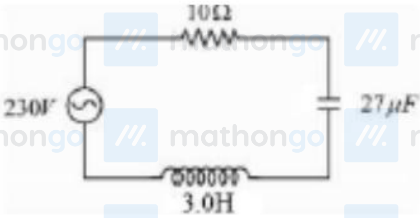
Q1 - 24 January - Shift 1

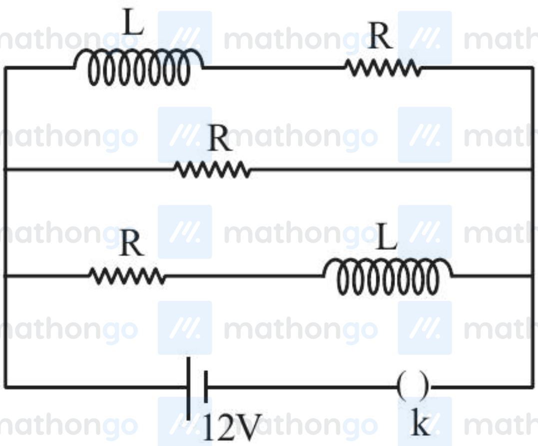
In the circuit shown in the figure, the ratio of the quality factor and the band width is ______ s.



Space for your notes:

Q2 - 24 January - Shift 2

Three identical resistors with resistance R = 12Ω and two identical inductors with sell inductance L=5 mH are connected to an ideal battery with emf of 12 V as shown in figure. The current through the battery long after the switch has been closed will be________A.



Space for your notes:

Q3 - 25 January - Shift 1

In an LC oscillator, if values of inductance and capacitance become twice and eight times, respectively, then the resonant frequency of oscillator becomes x times its initial resonant frequency ω_0. The value of x is:

(1) 1/4 (2) 16
(3) 1/16 (4) 4

Space for your notes:

Q4 - 25 January - Shift 1

An LCR series circuit of capacitance 62.5 nF and resistance of 50 Ω. is connected to an A.C. source of frequency 2.0 kHz. For maximum value of amplitude of current in circuit, the value of inductance is ______ mH.
(take $\pi^2 = 10$)

Space for your notes:

Q5 - 25 January - Shift 2

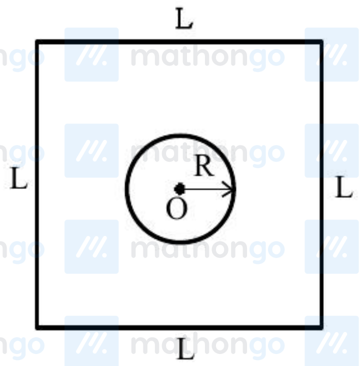
A series LCR circuit is connected to an AC source of 220 V, 50 Hz. The circuit contains a resistance $R = 80\Omega$, an inductor of inductive reactance $X_L = 70\ \Omega$, and a capacitor of capacitive reactance $X_C = 130\ \Omega$. The power factor of circuit is $\frac{x}{10}$.

The value of x is :

Space for your notes:

Q6 - 29 January - Shift 1

Find the mutual inductance in the arrangement, when a small circular loop of wire of radius 'R' is placed inside a large square loop of wire of side L (L >> R). The loops are coplanar and their centres coincide :



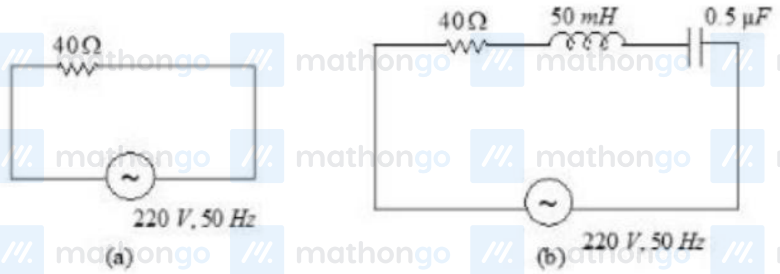
(1) $M = \frac{\sqrt{2}\mu_0 R^2}{L}$ (2) $M = \frac{2\sqrt{2}\mu_0 R}{L^2}$

(3) $M = \frac{2\sqrt{2}\mu_0 R^2}{L}$ (4) $M = \frac{\sqrt{2}\mu_0 R}{L^2}$

Space for your notes:

Q7 - 29 January - Shift 2

For the given figures, choose the correct options:



(1) The rms current in circuit (b) can never be larger than that in (a)

(2) The rms current in figure (a) is always equal to that in figure (b)

(3) The rms current in circuit (b) can be larger than that in (a)

(4) At resonance, current in (b) is less than that in (a)

Space for your notes:

Q8 - 29 January - Shift 2

An inductor of inductance 2 μH is connected in series with a resistance, a variable capacitor and an AC source of frequency 7 kHz. The value of capacitance for which maximum current is drawn into the circuit is $\frac{1}{x}$ F, where the value of x is______.

(Take $\pi = \frac{22}{7}$)

Space for your notes:

Q9 - 30 January - Shift 1

A sinusoidal carrier voltage is amplitude modulated. The resultant amplitude modulated wave has maximum and minimum amplitude of 120 V and 80 V respectively. The amplitude of each sideband is :

(1) 15 V
(2) 10 V
(3) 20 V
(4) 5V

Space for your notes:

Q10 - 30 January - Shift 1

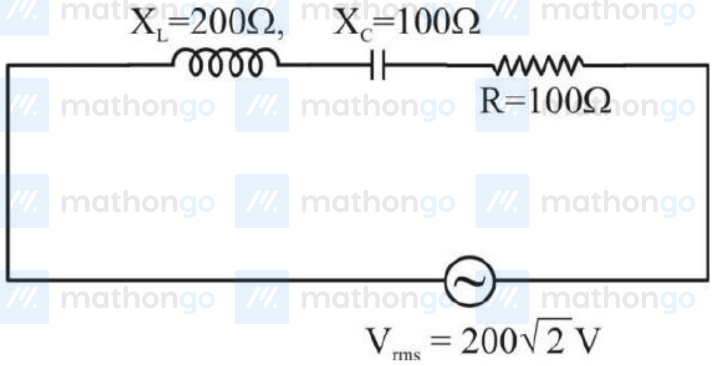
In a series LR circuit with $X_L = R$, power factor is P_1. If a capacitor of capacitance C with $X_C = X_L$ is added to the circuit the power factor becomes P_2. The ratio of P_1 to P_2 will be :

(1) $1:3$
(2) $1:\sqrt{2}$
(3) $1:1$
(4) $1:2$

Space for your notes:

Q11 - 30 January - Shift 2

In the given circuit, rms value of current (I_{rms}) through the resistor R is :



(1) 2A

(2) $\frac{1}{2}$A

(3) 20 A

(4) $2\sqrt{2}$A

Space for your notes:

Q12 - 30 January - Shift 2

In an ac generator, a rectangular coil of 100 turns each having area 14×10^{-2} m^2 is rotated at 360 rev/min about an axis perpendicular to a uniform magnetic field of magnitude 3.0 T. The maximum value of the emf produced will be ________ V. $\left(\text{Take } \pi = \frac{22}{7}\right)$

Space for your notes:

Q13 - 31 January - Shift 2

An alternating voltage source V = 260 sin (628t) is connected across a pure inductor of 5 mH. Inductive reactance in the circuit is:

(1) 3.14Ω

(2) 6.28Ω

(3) 0.5Ω

(4) 0.318Ω

Space for your notes:

Q14 - 31 January - Shift 2

A series LCR circuit consists of $R=80\Omega$ $X_L=100\Omega$ and $X_C=40\Omega$ The input voltage is $2500 \cos(100\,\pi t)$ V. The amplitude of current, in the circuit, is ________ A.

Space for your notes:

Q15 - 01 February - Shift 1

Match the List-I with List-II.

	List I		List II
A.	AC generator	I.	Presence of both L and C
B.	Transformer	II.	Electromagnetic Induction
C.	Resonance phenomenon to occur	III.	Quality factor
D.	Sharpness of resonance	IV.	Mutual Inductance

Choose the **correct** answer from the options given below:

(1) A-IV, B-II, C-I, D-III
(2) A-II, B-I, C-III, D-IV
(3) A-II, B-IV, C-I, D-III
(4) A-IV, B-III, C-I, D-II

Space for your notes:

Q16 - 01 February - Shift 1

A series LCR circuit is connected to an ac source of 220V, 50Hz. The circuit contain a resistance $R = 100\Omega$ and an inductor of inductive reactance $X_L = 79.6\,\Omega$. The capacitance of the capacitor needed to maximize the average rate at which energy is supplied will be _________ μF.

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (10)	**Q2** (3)	**Q3** (1)	**Q4** (100)
Q5 (8)	**Q6** (3)	**Q7** (1)	**Q8** (3872)
Q9 (2)	**Q10** (2)	**Q11** (1)	**Q12** (1584)
Q13 (1)	**Q14** (25)	**Q15** (3)	**Q16** (40)

Q1 (10)

$$\Delta\omega = \frac{R}{L}$$

$$Q = \frac{\omega_0}{\Delta\omega} = \omega_0 \frac{L}{R}$$

$$\omega_0 = \frac{1}{\sqrt{3\times 27\times 10^{-6}}} = \frac{1}{9\times 10^{-3}}$$

$$\frac{Q}{\Delta\omega} = \frac{\omega_0 \frac{L}{R}}{\frac{R}{L}} = \omega_0 \frac{L^2}{R^2} = \sqrt{\frac{1}{LC}}\frac{L^2}{R^2}$$

$$= \frac{1}{9\times 10^{-3}} \times \frac{9}{100} = 10\,\text{s}$$

Q2 (3)

After long time an inductor behaves as a resistance-less path.

So current through cell

$$I = \frac{12}{R/3} = 3\,\text{A}\ \{\because R = 12\Omega\}$$

Q3 (1)

The resonance frequency of LC oscillations circuit is

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$L \rightarrow 2L$

$C \rightarrow 8C$

$$\omega = \frac{1}{\sqrt{2L \times 8C}} = \frac{1}{4\sqrt{LC}}$$

$$\omega = \frac{\omega_0}{4}$$

So $x = \frac{1}{4}$

Q4 (100)

$$f = \frac{1}{2\pi\sqrt{LC}}$$

$$2000\,\text{Hz} = \frac{1}{2\pi\sqrt{L \times 62.5 \times 10^{-9}}}$$

$$L = \frac{1}{4\pi^2 \times 2000^2 \times 62.5 \times 10^{-9}} = 0.1\,\text{H} = 100\,\text{mH}$$

Q5 (8)

$$\cos\phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (X_C - X_L)^2}}$$

$$\cos\phi = \frac{80}{\sqrt{(80)^2 + (60)^2}}$$

$$\cos\phi = \frac{80}{100} \Rightarrow \frac{8}{10}$$

Q6 (3)

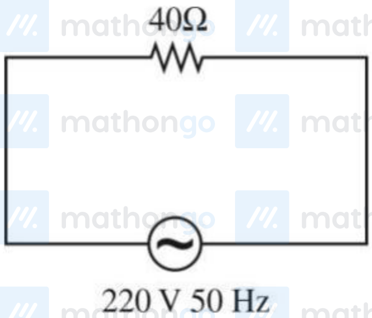
$$\phi = Mi$$

$$\phi = (\mathbf{B}A)$$

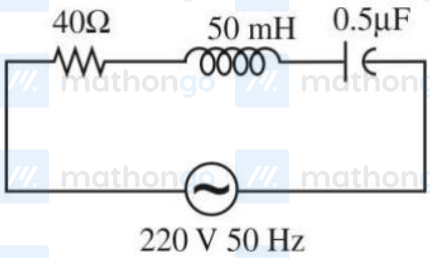
$$\phi = \pi R^2 \left(4 \frac{\mu_0}{4\pi} \frac{i}{\left(\frac{L}{2}\right)} \sqrt{2} \right)$$

$$\Rightarrow M = \frac{2\sqrt{2}\mu_0 R^2}{L}$$

Q7 (1)



$$I_{rms} = \frac{220}{40} = 5.5A$$



X_L is not equal to X_C. So rms current in (b) can never be larger than (a).

Q8 (3872)

$$\frac{1}{2\pi fC} = 2\pi fL$$

$$C = \frac{1}{4\pi^2 f^2 L} = \frac{1}{4 \times \pi^2 \times 49 \times 10^6 \times 2 \times 10^{-6}}$$

$$C = \frac{1}{3872} F$$

$x = 3872$ **Ans.**

Q9 (2)

$$A_c + A_m = 120$$
$$A_c - A_m = 80$$
$$\therefore \quad A_c = 100$$
$$A_m = 20$$

$$\text{Modulation index} = \frac{20}{100} = \frac{1}{5}$$

Amplitude of each sideband

$$= A_c \frac{(\text{mod ulation index})}{2}$$

$$= 100 \times \frac{1}{10} = 10 \text{ volt}$$

Q10 (2)

$$P = \frac{R}{Z} \Rightarrow P_1 = \frac{R}{\sqrt{R^2 + X_L^2}} = \frac{R}{R\sqrt{2}} \text{ (as } X_L = R)$$

$$P_1 = \frac{1}{\sqrt{2}}$$

$$P_2 = \frac{R}{\sqrt{R^2 + (X_L - X_L)^2}} = P_2 = 1$$

$$\frac{P_1}{} = \frac{1}{}$$

Q11 (1)

$$z = \sqrt{100^2 + (200 - 100)^2}$$

$$= 100\sqrt{2}\,\Omega$$

$$i_{rms} = \frac{V_{rms}}{z} = \frac{200\sqrt{2}}{100\sqrt{2}}$$

$$= 2A$$

Q12 (1584)

$$\xi_{max} = NAB\omega$$

$$= 100 \times 14 \times 10^{-2} \times 3 \times \frac{360 \times 2\pi}{60}$$

$$= 1584V$$

Q13 (1)

No Solution

Q14 (25)

No Solution

Q15 (3)

Based on theory

Q16 (40)

To maximize the average rate at which energy supplied i.e. power will be maximum.

So in LCR circuit power will be maximum at the condition of resonance and in resonance condition

$X_L = X_C$

$79.6 = \frac{1}{\omega C}$

$\therefore C = \frac{1}{2\pi \times 50 \times 79.6}$

$\therefore \boxed{C = 40 \mu F}$